

TP1 Test Cases

Team Project 1 — Input Validation Test Suite

These test cases follow the same semi-automated format used in `PasswordEvaluationTestingAutomation.java`: each case supplies an input and an expected pass/fail result, and the test is judged a success when the actual result matches the expected one (whether that means the input was correctly accepted or correctly rejected). Test cases for `UserNameRecognizer` and `PasswordEvaluator` are fully automatable, since neither class depends on any GUI widget. The Name/Email length checks are documented as manual test cases, since that logic is currently only reachable through the Update Account screen's dialogs.

1. UserNameRecognizer Test Cases

Automated via `UserNameRecognizer.checkForValidUserName(input)`. A test passes when an empty return string (valid) or a non-empty return string (invalid) matches the expected column.

#	Input	Expected	What It Tests
1	abcd	Pass (valid)	Minimum valid case: starts with a letter, meets 4-char minimum
2	abc	Fail (invalid)	Too short — below the 4-character minimum
3	"" (empty)	Fail (invalid)	Empty input is rejected immediately
4	1abcd	Fail (invalid)	Must start with a letter, not a digit
5	abcdefghijklmnop	Pass (valid)	Exactly 16 characters — boundary at max length
6	abcdefghijklmnopq	Fail (invalid)	17 characters — exceeds the 16-char maximum
7	ab.cd	Pass (valid)	Separator (.) followed by a valid letter
8	ab..cd	Fail (invalid)	Doubled separator — a separator cannot follow a separator
9	ab_cd	Pass (valid)	Underscore separator, followed by a valid letter
10	ab-cd	Pass (valid)	Hyphen separator, followed by a valid letter
11	ab&cd;	Pass (valid)	Ampersand separator, followed by a valid letter
12	abcd.	Fail (invalid)	Trailing separator with nothing valid following it
13	ab cd	Fail (invalid)	Space is not a permitted character anywhere in a username
14	Abcd1	Pass (valid)	Mixed case plus a digit in the body

2. PasswordEvaluator Test Cases

Automated via `PasswordEvaluator.evaluatePassword(input)`. Builds on the two example cases already present in the original `PasswordEvaluationTestingAutomation.java`, adding coverage for every individual requirement and the new maximum-length rule.

#	Input	Expected	What It Tests
1	Aa!15678	Pass (valid)	Satisfies every requirement (carried over from the original testbed)
2	A!	Fail (invalid)	Far too short; missing lower case and digit too
3	"" (empty)	Fail (invalid)	Empty input is rejected immediately

#	Input	Expected	What It Tests
4	alllowercase1!	Fail (invalid)	Missing an upper case letter
5	ALLUPPERCASE1!	Fail (invalid)	Missing a lower case letter
6	NoDigitsHere!	Fail (invalid)	Missing a numeric digit
7	NoSpecialChar1	Fail (invalid)	Missing a special character
8	Short1!	Fail (invalid)	7 characters — one below the 8-character minimum
9	Aa1!aaaa	Pass (valid)	Exactly 8 characters — boundary at minimum length
10	Aa1!aaaaaaaaaaaaaaaaaaaa aaaaaaaaaaaa	Fail (invalid)	33 characters — exceeds the 32-character maximum (added for TP1)
11	Aa1!aaaaaaaaaaaaaaaaaaaa aaaaaaaaaaaa	Pass (valid)	Exactly 32 characters — boundary at maximum length
12	Aa1!goodpwé	Fail (invalid)	Contains a character outside A-Z/a-z/0-9/special set

3. Name and Email Address Length Checks (Manual)

These fields (First Name, Middle Name, Last Name, Preferred First Name, Email Address) only enforce a 50-character maximum, checked in ViewUserUpdate before the database call. Tested manually via the Update Account screen, since the check is inline in a button handler rather than in a standalone class.

#	Input	Expected	What It Tests
1	"Jordan" (6 chars)	Pass (accepted)	Ordinary short name — well under the limit
2	50-character string	Pass (accepted)	Exactly at the limit — boundary case
3	51-character string	Fail (rejected)	One character over the limit — should show the validation alert
4	200-character string	Fail (rejected)	Simulates the "drop a huge string in the field" attack the assignment warns about

4. Status

Test cases 1–14 (Username) and 1–12 (Password) are implemented as runnable automated tests, following the same pattern as PasswordEvaluationTestingAutomation.java. Test cases in Section 3 have been verified manually against the running application.